

Worksheet 8 *Two populations*

Prepares you for lab manual pages **Chem1, Chem2** (section L3 S4)

Name:

Name of TA:

Suppose that A and B are populations of two species (aardvarks and baboons?) that obey the change equations

$$A' = 0.2 A - 0.004 A^2 + 0.03 AB, \quad B' = -0.1 B - 0.005 B^2 + 0.02 AB.$$

1. Draw a phase portrait for A if the population B equals zero (no baboons present). Shade in the circles or leave them empty, to indicate stable or unstable equilibria.

_____ → A

2. Draw a phase portrait for B if the population A equals zero (no aardvarks present). Shade in the circles or leave them empty, to indicate stable or unstable equilibria.

_____ → B

3. Does the aardvark population have a positive or negative effect on the baboon population? Does the baboon population have a positive or negative effect on the aardvark population? Write complete sentences, with reference to the appropriate interaction terms.

4. Challenge: are there any equilibrium points in which A and B are both positive? How can you interpret this, as a statement about the long-term behavior of these two populations?